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EXPERIMENTAL MEASUREMENT OF STEADY 3-D THERMAL GREEN'S FUNCTIONS

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ABSTRACT

In many areas of applied mathematics, the Green's function is the solution to a field problem due to the application of a potential or a gradient of a potential at a point or line in space on the solution domain. In heat transfer, the 'thermal' Green's function (TGF) is the temperature field that would arise as a result of the application of a point source of heat within or on the surface of a medium. We have been exploring the possible use of the thermal Green's function as a new way of characterizing the convective heat transfer from the surface of a solid that is cooled by a flow. An experiment was designed to explore possible ways in which the thermal Green's function can be measured in the laboratory. A small heated spot was created on the wall of a flow channel to approximate a point source of heat. The resulting temperature field was measured by the use of thermochromic liquid crystal thermography, and was compared to an analytical solution derived from the 3-D thermal Green's function. This paper reports progress towards understanding the limitations in creating and digitally photographing the temperature field from a small heat source on the wall of a convective flow.

Keywords: Thermochromic liquid crystal, Green's function, Forced convection

NOMENCLATURE

d brass rod diameter (m)
 h_{eff} effective heat transfer coefficient ($\text{W/m}^2 \text{K}$)
 h_{tip} heat transfer coefficient at the tip of the rod ($\text{W/m}^2 \text{K}$)
 h^* thermal conductance through the rod tip ($\text{W/m}^2 \text{K}$)
 H flow channel height (m)
 k brass rod thermal conductivity (W/m K)
 k_f effective fluid conductivity (W/m K)

k_{glass} glass cover slip thermal conductivity (W/m K)
 k_s solid conductivity (W/m K)
 L_{entry} channel entry length (m)
 m fin parameter ($1/\text{m}$)
 Nu average Nusselt number based on heat source
 Nu^* modified average Nusselt number based on heat source
 Pe Peclet Number
 Pe^* conjugate Peclet Number
 Pr Prandtl Number
 Re Reynolds Number
 Re_τ Reynolds Number based on the shear velocity
 t thickness of the glass cover slip (m)
 T measured surface temperature using TLC ($^{\circ}\text{C}$)
 T_{base} rod base temperature ($^{\circ}\text{C}$)
 T_{tip} rod tip temperature ($^{\circ}\text{C}$)
 T_{∞} local ambient temperature ($^{\circ}\text{C}$)
 $\bar{u}$ average velocity (m/s)
 u_τ shear velocity (m/s)
 W flow channel width (m)
 x spanwise direction
 y streamwise direction
 z normal direction

Greek Symbols

θ nondimensional temperature
 ν kinematic viscosity (m^2/s)
 ρ fluid density (kg/m^3)
 τ_w wall shear stress (Pa)

INTRODUCTION

The thermal Green's function is a fundamental method for describing convective heat transfer, perhaps the most fundamental. The Green's function approach uses point source solutions to the energy equation in order to construct the thermal response to arbitrary heat inputs. The utility of this methodology relies upon the ability of Green's function solutions to adequately represent the local flow dynamics, which establishes the need for accurate thermal Green's function measurements.

To the author's knowledge, the only previous attempt to measure the Green's function was the discrete Green's function approach of Batchelder and Eaton [1]. This method has shown considerable success for general 2-D steady flow measurements, but is limited in its ability to resolve true 3-D temperature fields.

This experiment is a first attempt to measure the Green's function in a canonical flow. A fully developed channel flow, in both laminar and turbulent regimes, was chosen because of its simplicity and the ability to characterize the near wall region as a linear shear flow. For the shear flow region, an analytical Green's function solution exists for negligible streamwise diffusion as demonstrated by Li and Ortega [2]. This solution is used to compute the temperature response to a steady prescribed heat flux distribution over a rectangular heat source. Li and Ortega [2,3] show that the Green's function approach yields the identical distribution of Nusselt number as a conventional laminar boundary layer unheated starting length approach [4].

The theoretical underpinnings for extracting the fundamental unsteady thermal Green's function by measuring the unsteady thermal response due to an impulsively started concentrated or spot source of heat on a surface is introduced in a companion paper [5]. The experimental challenge is to accurately resolve in space and time the temperature field on a surface due to that concentrated source.

As a first step, an experiment was designed to digitally image the steady color response of thermochromic liquid crystal (TLC) to a heated spot in the presence of a fully developed channel flow. The color response of the TLC is related to surface temperature through a previous calibration. The objective is to produce a well-controlled circular source of heat of known temperature, or known heat flux on the channel wall and assess our ability to resolve the resulting 2-D temperature field.

EXPERIMENTAL APPARATUS

The experimental measurement of steady 3-D thermal Green's functions (TGF) requires the measurement of the temperature field on a surface resulting from the application of a steady point source of heat. Since it is impossible to create a true point source, a theory is under development for extracting the TGF from a finite heat source [5]. This paper reports results from an experiment designed to measure such a

temperature response. Heat was supplied to one wall of 2-D channel flow by a circular source of diameter 0.165 cm (0.065"). Thermochromic liquid crystal (TLC) was used to view the temperature response and a calibration was utilized to relate TLC color to temperature. Two channel flow cases were investigated, one laminar and one turbulent, and the results compared.

The experimental apparatus consists of two main parts: the flow channel and the test section. In the experiment the flow channel height, H , is 0.635 cm (0.250"). The channel, shown in Figure 1.a, is constructed of flat and smooth aluminum tooling plate to minimize flow disturbances. The channel width, W , is 30.48 cm (12.0"), and the aspect ratio W/H is 48. The large W/H ratio minimizes sidewall effects in the center region where the test section is located. Airflow is provided by compressed air or a high head centrifugal blower with a variable frequency drive (VFD), which allows the output of the blower to be accurately controlled. The air is passed through a diffuser, a flow management section, and a 24:1 contraction before it enters the flow channel. The long channel entry length, L_{entry}/H of 220, assures that the flow is fully developed at the test section position.

The test section consists of a wooden box containing an electrical heater, a copper heat intensifier, and a brass rod with diameter, d , of 0.165 cm (0.065"), Figure 1.b. The heat source is produced by a 2.54 cm x 2.54 cm, 70 Ω electrical resistance heater. The heater is in contact with the base of a conical shaped copper heat intensifier, which conducts the heat to the base of the brass rod. The rod conducts the heat to the bottom of the thin glass plate. The wooden box is filled with polystyrene foam bead insulation to reduce heat losses as shown in Figure 1.b. At the surface of the flow channel a 2.223 cm (0.875") x 2.223 cm (0.875") glass cover slip, 0.018 cm (0.007") thick, is bonded to the foam substrate with a small air gap enclosed between the foam and the glass in an effort to minimize the conjugate conduction effects in the substrate. A physical constraint on the cover slip is that it must be rigid, thin, smooth and able to be painted. The combination of an insulating air layer and the thin glass layer greatly reduced, but did not eliminate, conjugate conduction effects.

The flow side of the glass cover slip is coated with flat black paint and then thermochromic liquid crystal with a nominal temperature bandwidth of 30-35 °C. The bandwidth is a characteristic of the TLC and refers to temperature range in which the TLC undergoes a color change. A Plexiglas window in the top wall of the flow channel allowed the color response of the TLC to be digitally imaged. Calibration of the liquid crystal was performed in a separate experiment using a thermoelectric cooler based calibrator. Calibration was performed by viewing through a Plexiglas window, under the same lighting conditions as used in the experiment. It was found that the presence of the Plexiglas window changed the calibration very little. The digital images were captured using a Sony DFW-VL500 digital camera and a National Instruments LabVIEW™ based imaging software developed by the Experimental and

Computational Heat Transfer Group at the University of Arizona [6]. The software uses a TLC calibration in order to relate the hue value of each pixel to surface temperature.

TLC Calibration

Proper calibration of the TLC is essential to resolving the surface temperature distribution on the substrate. Hue value and temperature are related using the calibration shown in Figure 3. The calibration was obtained using the same set up as the experiment. The digital camera images the calibration surface, which is held at a constant temperature by a thermoelectric cooler (TEC), and the hue value is recorded as a function of surface temperature measured with a calibrated thermistor.

The useable range of the hue temperature calibration curve limits the usable range of hue values from the images recorded by the digital camera. From Figure 3 it is concluded that the slope of the hue vs. temperature plot is the greatest in the range of temperature values between 30.5-30.75 °C. This limits the useable range of calibration to hue values corresponding to temperatures of 30.5-30.75 °C. Using this limited range of the calibration provides the highest sensitivity to changes in surface temperature and, therefore greater accuracy in measuring the temperature.

Estimation of Rod Tip Temperature

The heat transfer model for the rod used to conduct heat to the channel surface is illustrated in Figure 4. The purpose of the model is to predict the heat source average temperature by proper accounting of heat losses from the rod. The rod is modeled as a pin fin and the tip temperature is calculated as

$$\frac{d^2\theta}{dx^2} - m^2\theta = 0, \quad (1)$$

$$\text{where } \theta = T(x) - T_\infty, \quad m^2 = \frac{4h_{eff}}{kd}.$$

Boundary conditions are:

$$x = 0 \quad \theta = T_{base} - T_\infty \quad (2)$$

$$x = L \quad -k \frac{d\theta}{dx} = h^* \theta \quad (3)$$

$$\text{Let } h^* = \left(\frac{t}{k_{glass}} + \frac{L}{h_{tip}} \right)^{-1}. \quad (4)$$

The solution for the tip temperature T_{tip} is given by:

$$\frac{T_{tip} - T_\infty}{T_{base} - T_\infty} = \cosh(mL) - \frac{h^* \cosh(mL) + km \sinh(mL)}{h^* \sinh(mL) + km \cosh(mL)} \sinh(mL). \quad (5)$$

The rod thermal model showed that the heat losses from the sides of the rod were minimal and therefore the tip temperature was mildly sensitive only to the tip conductance given by Equation 4. For these canonical shear flows, it was

possible to obtain a very good estimation of h^* , hence, the uncertainty in T_{tip} was low.

RESULTS

Two flow cases were investigated and compared, laminar channel flow and turbulent channel flow. In each case, Reynolds numbers are used to characterize both global flow behavior and local flow behavior.

$$Re = \frac{\bar{u} 2h}{\nu} \quad (6)$$

$$Re_\tau = \frac{u_\tau d}{\nu} \quad (7)$$

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}} \quad (8)$$

For the global flow, a Reynolds number based on the mean flow velocity and the hydraulic diameter of the channel is defined in Equation 6, while a Reynolds number based on the shear velocity and the heat source diameter, as defined in Equation 7, effectively characterizes the local flow dynamics of the shear flow model [5]. The shear velocity is related to the local shear stress at the wall using Equation 8.

In the experimental results for both flow cases considered the influence of conduction in the glass cover slip is apparent. This is as a result of conduction effects within the glass being of the same order as the convection [2,3]. The conjugate Peclet number is defined as,

$$Pe^* = \frac{k_f d}{k_s t} Pe^{2/3} \quad (9)$$

with,

$$Pe = Re_\tau Pr \quad (10)$$

$$Nu^* = \frac{k_f d}{k_s t} Nu \quad (11)$$

$$Nu = \frac{h_{tip} d}{k_s}. \quad (12)$$

Li and Ortega [2,3] have shown that the conjugate Peclet number for thin plates is the characteristic indicator of the importance of conjugate conduction effects. If Pe^* is low, conjugate effects are non-negligible. From Figure 5, abstracted from Li and Ortega [2], Pe^* calculated for the present experiments is of order 3.0 indicating that despite the measures taken to minimize substrate conduction, conjugate effects are significant.

The most meaningful results for temperature can be shown by nondimensionalizing the temperature as,

$$\theta = \frac{T - T_\infty}{T_{tip} - T_\infty}. \quad (13)$$

Data taken for different tip temperatures at the same shear flow Reynolds number, Re_τ , can be superimposed on one plot of nondimensional temperature. One isotherm from each set of data at a specific tip temperature, T_{tip} , is chosen in the range of

temperature between 30.5-30.75 °C. The nondimensional temperature plots are shown for the laminar and turbulent flow cases in Figures 6 and 7.

DISCUSSION

The results for the laminar flow case, with $Re = 1000$ and $Re_\tau = 14$, are shown in Figure 6. The data show a wide thermal plume, which indicates the presence of lateral conduction effects. This is attributed to the low conjugate Peclet number, $Pe^* = 0.80$. The black circle in the center of the figure represents the size of the heat source and the x and y axis scales are in dimensions of pixels. This dimension represents the resolution of the image taken by the digital camera used to acquire the temperature data. At each (x, y) pixel location a value of temperature is found from the calibration. In Figure 6 the three isotherm lines correspond to three tip temperatures and as the T_{tip} increases the isotherm lines move further from the source as expected.

Results for the turbulent channel flow case are shown in Figure 7. The corresponding Reynolds numbers for the turbulent channel flow case are $Re = 11500$ and $Re_\tau = 98.7$, and the conjugate Peclet number, $Pe^* = 2.95$. As a result of the higher conjugate Peclet number in the turbulent channel flow case the effects of conjugate conduction in the glass cover slip are reduced, however they are still present. This is visible when comparing Figures 6 and 7, the isotherm lines in Figure 6 demonstrate greater lateral spreading than those in Figure 7. The isotherm lines in Figure 7 are more convectively dominated.

An analytic Green's function is available for comparison to the measured data, as used by Li [3]. This solution neglects streamwise diffusion, but gives reasonable results for high Re . In order to compare with measured data, this Green's function was integrated over a circular heat source to obtain nondimensional temperature contours. In contrast to the experiment, which can be modeled as a constant temperature source, this solution corresponds to a constant heat flux distribution. Although the boundary conditions differ, the results can be used for a qualitative comparison. The x and y coordinate axes for the analytical solutions are nondimensionalized with respect to the radius of the heat source.

The solution using the analytical Green's function for the laminar flow case is shown in Figure 8. The comparison between the experimental data shown in Figure 6 and analytical results Figure 8 show a large difference in the heat spreading in the spanwise and upstream directions, while the downstream convection effects appear to be of similar order. This can be attributed to the low conjugate Peclet number, Pe^* in the experiment.

In the turbulent channel flow case, the shear flow region is much thinner. The thermal boundary layer in the normal direction for the current spot size exceeds the linear shear flow region of the fluid so that the shear flow model cannot be used

in the turbulent flow case. Nevertheless, the experimental results for turbulent channel flow cases are shown in this paper for illustration, Figure 7.

The solution using the analytical Green's function for the turbulent flow case is shown in Figure 9. The comparison of the experimental data in Figure 7 and the analytical solution clearly shows the presence of conjugate conduction in the experimental data. The analytical solution shows no conduction in the spanwise and upstream directions, while the experimental data shows considerable conjugate effects. In addition, the experimental results show much weaker convection than was predicted using the analytical thermal Green's function.

From the data shown in Figure 5 it can be concluded that for small conjugate Peclet numbers conduction in the substrate will play an important role in this experiment. Efforts have been made to minimize conjugate conduction effects in the surface by maximizing Pe^* by using a very thin glass cover slip with an insulating air layer beneath it. Reducing the thickness of the cover slip from 0.018 cm (0.007") would increase Pe^* , but may compromise the structural integrity of the cover slip. Another way to increase Pe^* is to increase Re_τ , which increases Pe^* to the 2/3 power of Re_τ thereby providing only a mild benefit to Pe^* . The conductivity ratio of k_f / k_s could be increased, however, the limitations on materials for the cover slip fix the conductivity of the solid. The diameter of the rod could be increased, however, this would compromise the ability to keep the thermal boundary layer within the linear shear region.

CONCLUSIONS AND RECOMMENDATIONS

1. The results of this investigation attest to the difficulty in creating a heated spot where conjugate conduction is not a factor.
2. The direct measurement of tip temperature, T_{tip} , was not feasible using the current technique because a wider bandwidth liquid crystal would be required. Alternative heat sources such as impinging lasers would have similar problems.
3. The experiment has shown that digital imaging of the temperature response of TLC is practical, accurate and valuable.
4. Digital photography is easier to use than film and the images are instantly viewable. In this experiment the physical size of the temperature field was small so that the resolution of the camera was an issue. However, zoom optics and currently available cameras with much greater resolution are more than sufficient to overcome issues of spatial resolution for even smaller spot sizes.

FIGURES

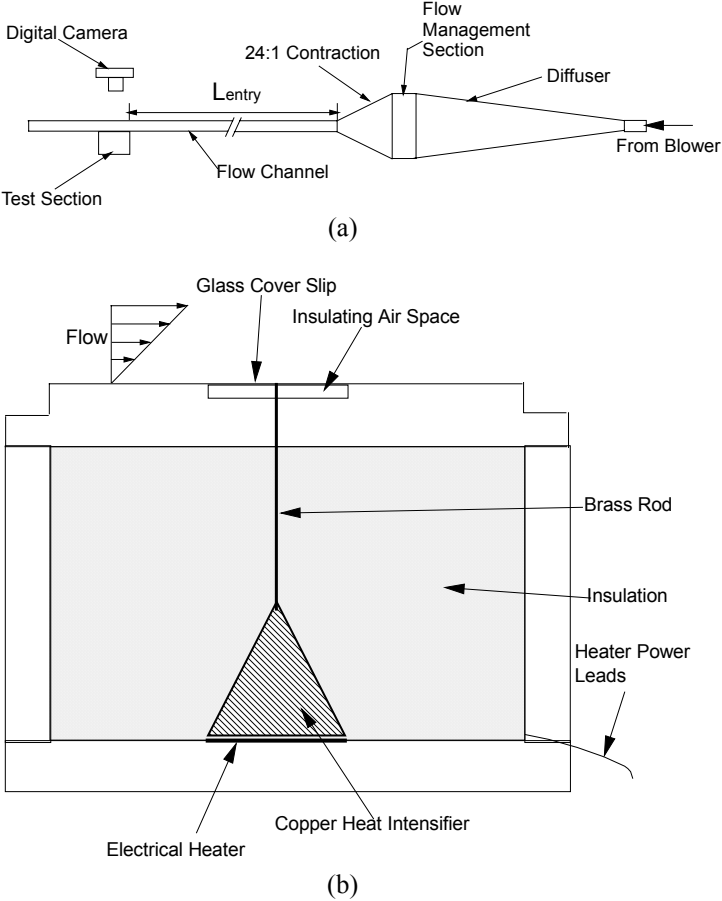


Figure 1, Experimental Apparatus; (a) Flow Channel, (b) Test Section

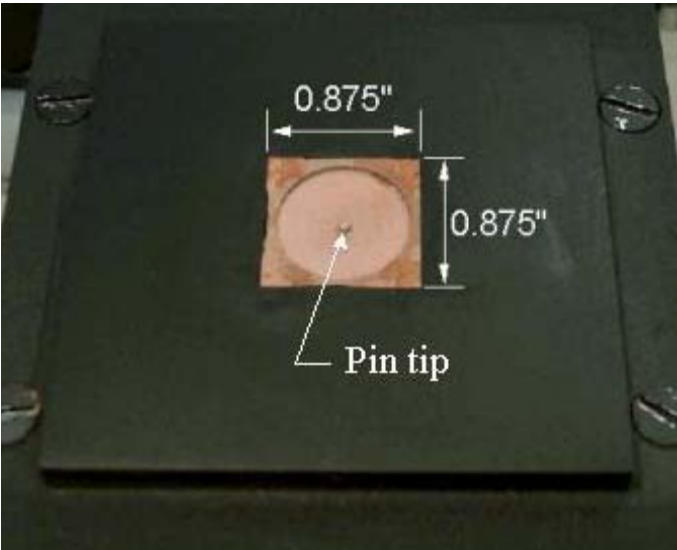


Figure 2, Channel Wall Heat Source with Glass Cover Slip Removed

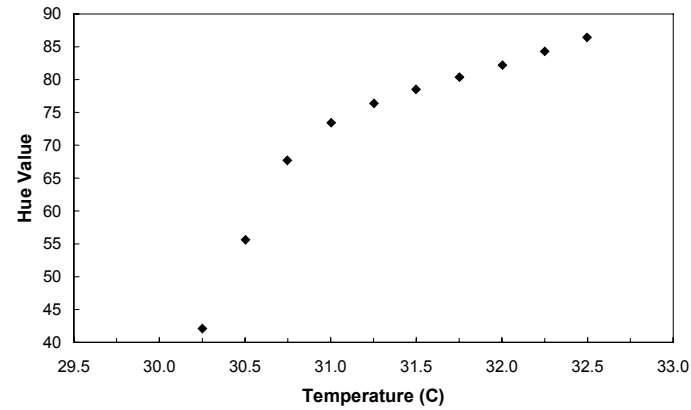


Figure 3, Typical Hue vs. Temperature Calibration Curve

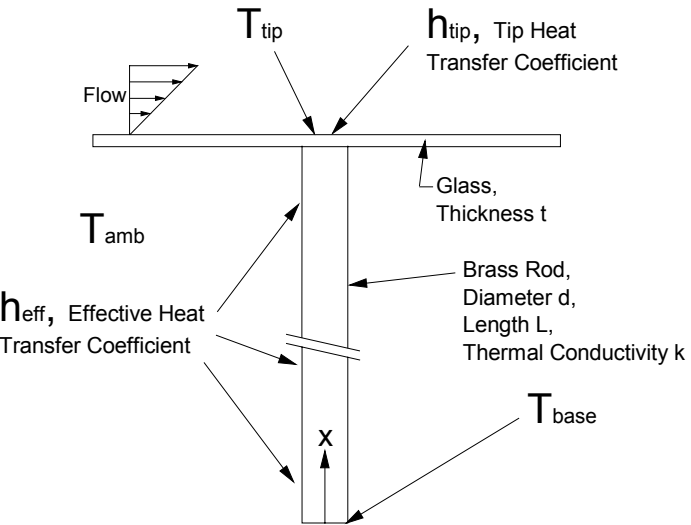


Figure 4, Tip Temperature Model

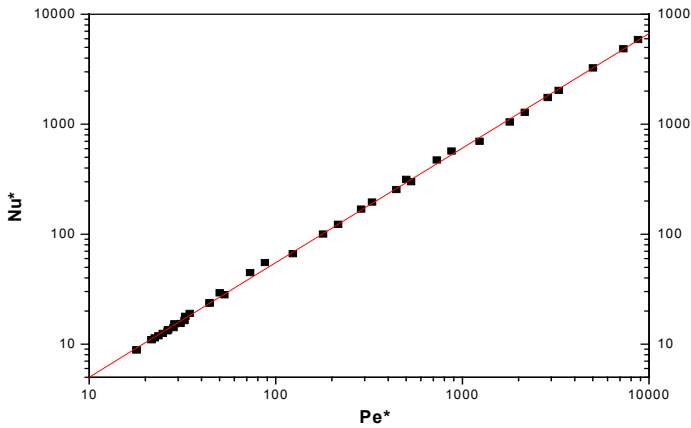


Figure 5, Modified averaged Nu^* over the rectangular source (1cmx2cm) and square source (2cmx2cm) vs. conjugate Pe^* number from Li and Ortega [2].

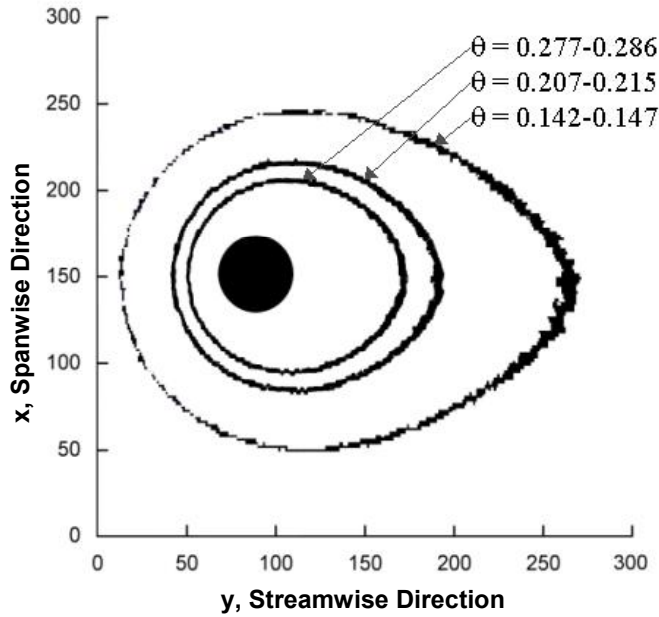


Figure 6, Laminar Channel Flow Nondimensional Temperature Contour Plot for Different Rod Tip Temperatures, T_{tip} , x and y scales are in pixels

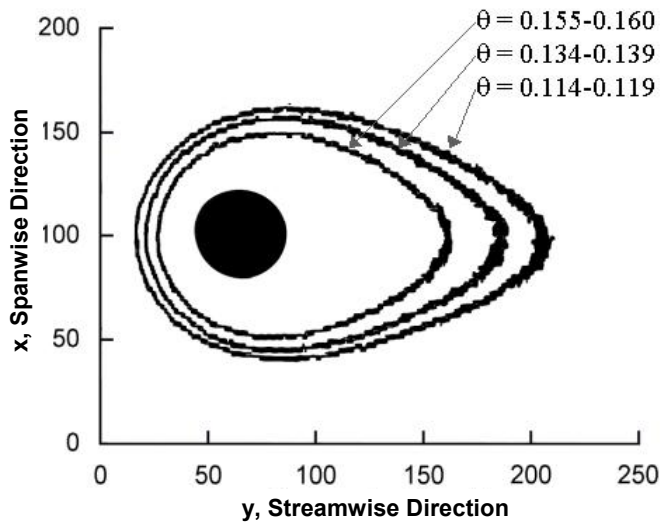


Figure 7, Turbulent Channel Flow Nondimensional Temperature Contour Plot for Different Rod Tip Temperatures, T_{tip} , x and y scales are in pixels

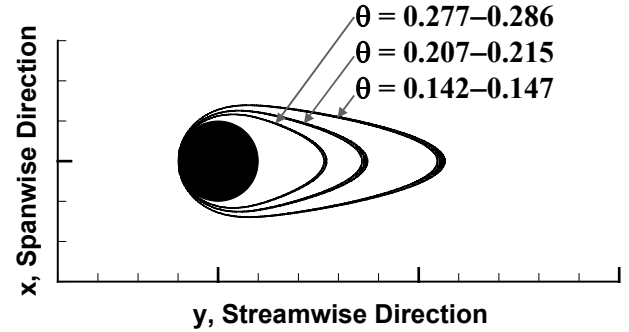


Figure 8, Laminar Flow Case, Numerical Data Obtained using an Analytical Thermal Green's Function

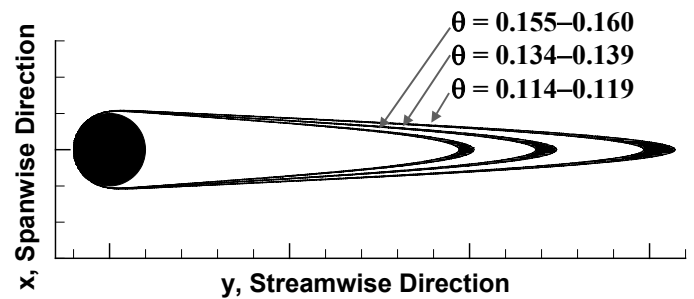


Figure 9, Turbulent Flow Case, Numerical Data Obtained Using an Analytical Thermal Green's Function

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